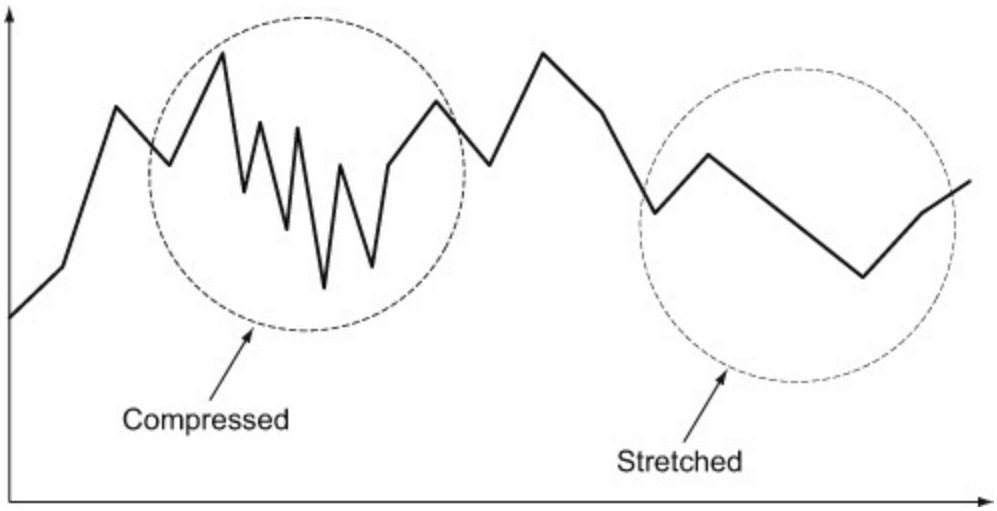
event loop by executing a long-running process. This is a problem for any program, but it's *visibly* bad for your animations. Figure 11.5 shows what happens when the event loop is blocked for something time-critical like drawing a graph.

Figure 11.5. Some `setInterval` inaccuracy is making the graph inaccurate.



If you block the event loop for 5 seconds, then instead of the animations you want, you'll get incorrect (and often surprising) animations. The same problem happens for any time-sensitive drawing inside a browser. Remember, the browser provides you with a single event loop—anything that happens on the event loop can impact time-based rendering.

The issue of timeout accuracy has more subtle implications. Although you'll notice if your event loop is blocked for a second, you won't normally notice if your event loop is blocked for 50 milliseconds. For an animation, even a 50-millisecond inaccuracy can make it look terrible. For something time-critical like a game, accuracy of the timing and animations is essential.

One tactic for getting around this is called *drift compensation*: